

7 (a) line has non-zero intercept/line does not pass through origin

B1

charge is/should be proportional to potential (difference)

or

charge is/should be zero when p.d. is zero

(therefore there is a systematic error)

B1

[2]

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(b) reasonable attempt at line of best fit

B1

use of gradient of line of best fit clear

M1

$C = 2800 \mu\text{F}$ (allow $\pm 200 \mu\text{F}$)

A1 [3]

(c) energy = $\frac{1}{2} CV^2$ or energy = $\frac{1}{2} QV$ and $C = Q / V$

C1

$\Delta \text{ energy} = \frac{1}{2} \times 2800 \times 10^{-6} \times (9.0^2 - 6.0^2)$

C1

$= 6.3 \times 10^{-2} \text{ J}$

A1 [3]